

Invited Review

Quaternary palaeoenvironmental archives of the Indian subcontinent and their imprint on herpetofaunal biogeography: A multi-proxy synthesis

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ABSTRACT

This review presents the first integrated proxy, mechanism and biotic-response framework for the Indian Quaternary, linking the Subcontinent's dense network of palaeoenvironmental archives to the biogeographic trajectory of its endemic herpetofauna. Speleothem $\delta^{18}\text{O}$ records, Arabian Sea and Bay of Bengal sediment cores, Karewa and peninsular lake sequences, Siwalik and Narmada alluvium, and Kurnool cave deposits together document Indian Summer Monsoon (ISM) variability on orbital, millennial, centennial and historical timescales (the historical band here denoting the past ~2000 years, resolved by dated speleothem and documentary records). Against this proxy backbone, the fossil and molecular phylogeographic record of Indian amphibians and reptiles is evaluated as a biogenic assemblage here, the body of biological evidence (fossils, subfossils and the phylogeographic signatures of living taxa) that, like pollen or foraminifera, records past environmental conditions that complements geological and climatological archives. Three patterns emerge. At orbital scale, ISM variability aligns with Plio-Pleistocene diversification slowdowns in Western Ghats ranoids and with the Late Pliocene lineage divergence of the Himalayan toad *Duttaphrynus himalayanus* across the Thakkhola basin. Abrupt Holocene excursions, including the 8.2 ka and 4.2 ka events, are resolved at high precision in speleothem and marine archives but remain untested against coalescent demographic inference in Indian herpetofauna. Late Pleistocene aridity, most severe at the Last Glacial Maximum, coincides with community-level impoverishment in peninsular cave faunas at Kurnool, although the available evidence establishes co-occurrence rather than a demonstrated causal role for aridity relative to co-varying thermal and vegetation change. Three proxy-calibrated research priorities follow: microvertebrate taphonomy in Narmada and Karewa sequences, population genomics tied to speleothem chronology, and species-distribution hindcasting to the LGM, mid-Holocene and 4.2 ka intervals. The synthesis contributes to a biogenic assessment of Quaternary-to-Recent climate impacts on life on the Indian Subcontinent, aligned with UN Sustainable Development Goals 13 (Climate Action) and 15 (Life on Land), and supports future projections of amphibian and reptile response to ongoing change.

1. Introduction

The Indian Subcontinent (here including India, Pakistan, Nepal, Bhutan, Bangladesh and Sri Lanka) preserves one of the densest networks of Quaternary palaeoenvironmental archives on Earth (Prasad and Enzel, 2006; Banerji et al., 2020). Cave carbonates of northern and peninsular India, deep-sea sediment cores from the Arabian Sea and Bay of Bengal, glaciolacustrine successions in the Kashmir Valley, fluvial alluvium of the Narmada, Ganga and Godavari valleys, and cave fills at Kurnool span a record of Indian Summer Monsoon (ISM) variability from orbital to annual timescales. These archives have transformed understanding of the South Asian monsoon system, its global teleconnections, and its interactions with human and non-human biota

(Gupta et al., 2003; Sinha et al., 2007, 2015; Kathayat et al., 2017).

Biotic responses to Quaternary environmental change on the Subcontinent have been examined unevenly. Mammalian palaeontology of the Siwalik and Narmada sequences has a long tradition (Patnaik, 2003; Patnaik et al., 2009), and pollen archives supply the vegetation context (Dodia et al., 1984; Agrawal et al., 1989; Sharma et al., 2004). Indian amphibians and reptiles, despite constituting one of the world's most endemic herpetofaunal assemblages (Myers et al., 2000; Dinesh et al., 2009; Dahanukar and Molur, 2020), have rarely been used as systematic biological proxies for palaeoenvironmental reconstruction, in contrast to the long-established use of pollen and vertebrate (chiefly mammalian) fossils. As ectotherms with narrow physiological tolerances and strong dependence on hydrology and vegetation, herpetofauna

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constitute, in principle, a sensitive biotic archive. Contemporary conservation assessments reinforce this. The second Global Amphibian Assessment concludes that amphibians remain the most threatened vertebrate class (40.7% of species globally threatened), with climate change now driving 39% of status deteriorations since 2004 and the Western Ghats–Sri Lanka among the most affected regions (Luedtke et al., 2023). For reptiles, the first global reptile assessment likewise found at least 21.1% of species to be threatened with extinction, with agriculture, logging, urban development and invasive species as the principal drivers and forest-dwelling species most at risk (Cox et al., 2022); South Asia, with its high squamate and chelonian diversity, is prominent among the regions of conservation concern. A Quaternary perspective is therefore directly relevant to projecting how the geographic distributions and population sizes of Indian amphibians and reptiles may shift under twenty-first-century climate and land-use change.

This review has two aims. The first is to synthesise Quaternary palaeoenvironmental archives of the Indian Subcontinent (speleothem, marine, lacustrine, fluvial and cave-deposit) within a proxy, mechanism and response framework appropriate to a multi-proxy assessment. The second is to use the Subcontinent's herpetofauna, including its fossil record and its molecular phylogeographic record, as a biogenic assemblage through which the effects of Quaternary-to-Recent climate and environmental change on life on Earth can be read at the regional scale of the Indian Subcontinent. The geographic distribution of the archives and of the biogeographic regions treated here is summarised in Fig. 1; the conceptual logic of the framework is summarised in Fig. 2. The structure is as follows. Section 2 sets out methodology. Section 3 reviews the proxy archives, organised by archive class. Section 4 treats the Quaternary herpetofaunal fossil record with attention to taphonomy. Section 5 surveys the phylogeographic evidence. Section 6 integrates the two literatures in a framework table that links proxy, signal, biotic mechanism and response. Section 7 places these results in the context of Anthropocene projections, and Section 8 sets research priorities.

2. Materials and methods

2.1. Scope and temporal focus

Temporal scope is the Quaternary (~2.58 Ma to present), with emphasis on the Late Pleistocene, Holocene and historical periods where proxy resolution is highest. Geographic scope is the Indian Subcontinent and the adjoining northern Arabian Sea and Bay of Bengal that record its terrestrial runoff and monsoon-driven upwelling (Gupta et al., 2003; Saraswat et al., 2013). A compressed Neogene pre-Quaternary baseline (Section 5.1) is retained because the principal radiations of Indian herpetofauna predate the Quaternary and must be acknowledged before intra-Quaternary signals can be interpreted.

2.2. Literature retrieval

Two parallel corpora were assembled. The palaeoenvironmental corpus was searched in Web of Science and Scopus with combined terms ("Indian Summer Monsoon" OR "ISM" OR "Asian monsoon" OR "palaeoclimate") AND ("Holocene" OR "Pleistocene" OR "Quaternary") AND ("speleothem" OR " $\delta^{18}\text{O}$ " OR "foraminifer" OR "pollen" OR "diatom" OR "Mg/Ca" OR "Narmada" OR "Siwalik" OR "Karewa" OR "Kurnool" OR "Arabian Sea" OR "Bay of Bengal" OR "upwelling") for the period 1990–2025. Primary data papers were privileged over reviews; where a review was used (Banerji et al., 2020; Sinha et al., 2018), primary references were back-traced. The herpetofaunal corpus combined keyword searches for ("India" OR "Sri Lanka" OR "Western Ghats" OR "Himalaya" OR "phylogeography" OR "molecular phylogeny" OR "diversification" OR "amphibian" OR "frog" OR "caecilian" OR "reptile" OR "squamate" OR "turtles"), together with the Indian herpetological fossil literature (Siwaliks, Narmada, Kurnool) and recent taxonomic revisions of fossil crocodyloids (Rio and Mannion, 2024) and amphibian phylogeography (Hofmann et al., 2026). Records returned by both searches were screened on title and abstract and retained when they met three criteria:

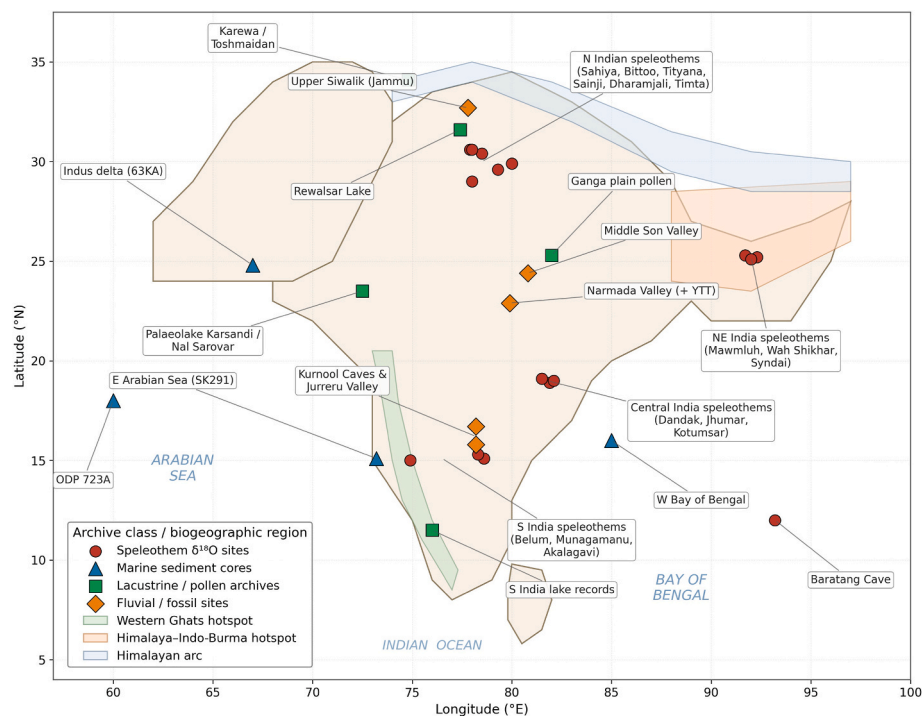


Fig. 1. Distribution of Quaternary palaeoenvironmental proxy sites on the Indian Subcontinent, classified by archive type (speleothem $\delta^{18}\text{O}$, marine sediment cores, lacustrine or pollen sequences, fluvial or cave fossil sites), with overlay of the Western Ghats and Himalaya–Indo-Burma biodiversity hotspots and the Himalayan arc. Outlines are schematic. Fig. 1 is an original compilation by the author; proxy-site locations were compiled from the primary studies cited in Section 3 and Table 1, and the base map and hotspot outlines are schematic rather than georeferenced. Proxy-site names and principal citations are given in Section 3 and Table 1.

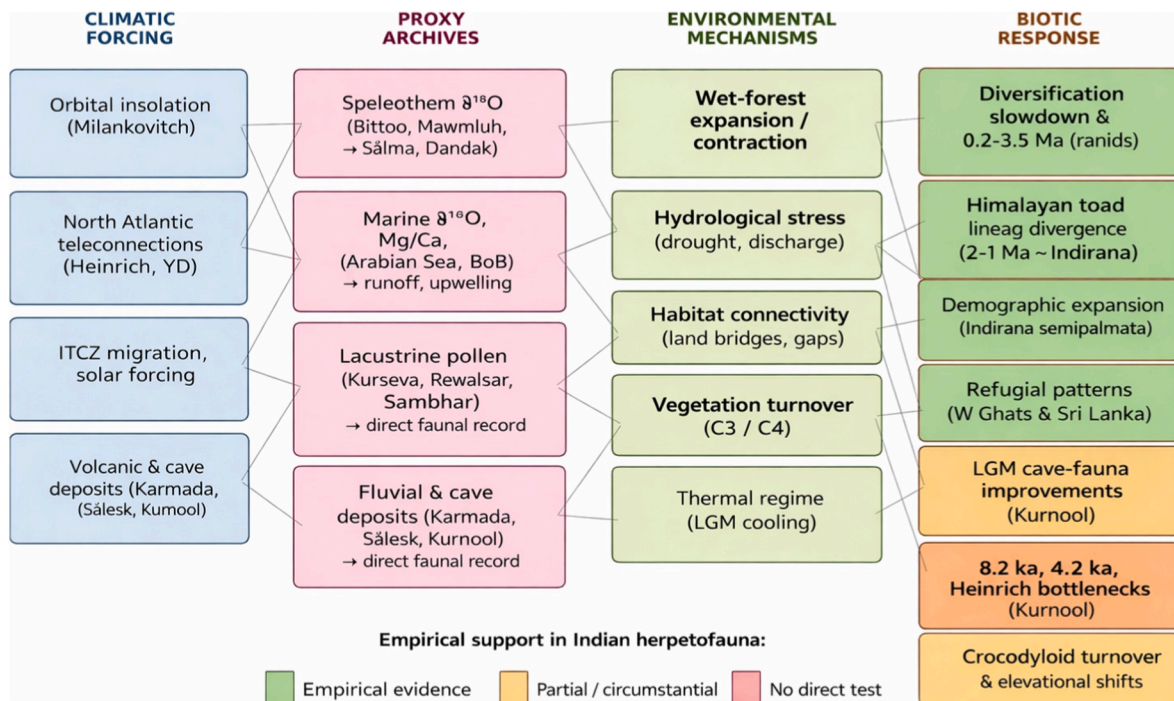


Fig. 2. Conceptual framework linking Quaternary palaeoenvironmental archives of the Indian Subcontinent to herpetofaunal response. Climatic forcings (orbital, teleconnection-driven, volcanic and tectonic) are recorded in four classes of proxy archive, which capture environmental mechanisms (hydrological regime, vegetation, connectivity, thermal regime) through which biotic response is mediated. Colour coding of the biotic-response column indicates empirical support in Indian herpetofauna: green for direct empirical evidence, amber for partial or circumstantial, red for no direct test.

(i) they reported a primary, or well-supported synthetic, proxy or phylogeographic dataset; (ii) they provided explicit chronological control (radiometric, magnetostratigraphic, tephrostratigraphic or molecular-clock); and (iii) they were geographically relevant to the Indian Subcontinent as defined above. Studies that were purely methodological, or that lay outside South Asia, were consulted only for conceptual or comparative context. This review is deliberately selective rather than exhaustive and is not a systematic (PRISMA-type) review: it is organised around the strongest available record in each archive class, and the corpus synthesised here is represented by the studies cited in Sections 3–5 and compiled in Table 1. Because the selection privileges well-dated, high-resolution records, it may under-represent lower-resolution or unpublished archives; this bias is stated explicitly so that it can be weighed.

2.3. Integration strategy

The two corpora are integrated under a proxy, signal, mechanism and biotic-response chain (Fig. 2). For each proxy record, the framework specifies (a) the climatic or environmental variable it tracks and its temporal resolution, (b) the causal pathway through which that variable would be expected to affect amphibians and reptiles (thermal regime, hydrological regime, vegetation structure, connectivity), and (c) the empirical status of the expected response in Indian herpetofauna. The chain is operationalised in Table 1 (Section 6), so that empirical coverage is explicit and research gaps are identifiable. Entries in Table 1 are colour-coded according to empirical support.

2.4. Taphonomic and preservation considerations

The Indian Quaternary herpetofossil record is sparse relative to mammals for three convergent reasons. Preservation is a first bias: small, porous amphibian and small-lizard bone dissolves in acidic sediment and fractures in fluvial transport (Behrensmeyer, 1978; Lyman, 1994). Collection is a second: historical excavations at Siwalik, Narmada and

Kurnool sites privileged megafaunal material and did not systematically screen-wash for microvertebrates (Badam, 1979; Patnaik et al., 2008, 2009). Chronology is a third: many faunal assemblages lack direct dating of the herpetofaunal fraction, limiting correlation to precisely dated monsoon events (Patnaik et al., 2008, 2009; Turvey et al., 2021). Throughout this review, absence from published assemblages is therefore interpreted with caution: it may reflect collection or preservation bias, genuine past absence, or a combination of these, and the alternatives are distinguished only where independent stratigraphic, taphonomic or ecological evidence allows.

3. Quaternary palaeoenvironmental archives of the Indian subcontinent

3.1. Speleothem $\delta^{18}\text{O}$ records of ISM variability

Indian speleothems provide the most precisely dated, highest-resolution Quaternary climate records available from the Subcontinent. Stalagmite $\delta^{18}\text{O}$ records the isotopic composition of monsoon moisture, which over the Indian monsoon domain is dominated by the amount effect: more negative $\delta^{18}\text{O}$ reflects stronger ISM precipitation (Sinha et al., 2007). A distributed network now resolves ISM variability from orbital to sub-annual timescales (Fig. 1). In the north, Bittoo, Sahiya, Tityana and Sainji caves (Sinha et al., 2011, 2015; Kathayat et al., 2017) span the last ~240 kyr. Dharamjali and Timta in the central Himalaya provide complementary Late Pleistocene records. In northeast India, Mawmluh, Syndai and Wah Shikhar caves deliver annually resolved Holocene records at the Meghalaya Plateau (Berkelhammer et al., 2012; Kathayat et al., 2018). In the core monsoon zone and peninsular south, Dandak, Jhumar, Kotumsar, Valmiki, Munagamanu, Nakarallu and Akalagavi close the spatial network (Sinha et al., 2007; Dutt et al., 2015).

Three scales of variability are relevant. At orbital scale, ISM strength tracks Northern Hemisphere summer insolation across glacial–interglacial cycles (Kathayat et al., 2017), with interglacial maxima

Table 1

Proxy, mechanism and response framework linking Quaternary palaeoenvironmental archives of the Indian Subcontinent to herpetofaunal response. Geographic scope of each proxy record is indicated in brackets.

*Note. Colour coding: green for direct empirical evidence linking proxy to Indian herpetofaunal response; amber for partial or circumstantial support; red for no direct published test. Red rows identify research priorities rather than demonstrated biological non-response. Geographic scope tags in column 1 indicate the region of the Indian Subcontinent to which each proxy archive applies.

Proxy record [scope] (key studies)	Environmental signal (timing, magnitude)	Expected herpetofaunal mechanism	Empirical evidence in Indian herpetofauna
Speleothem $\delta^{18}\text{O}$ networks [Indian Subcontinent] (Sinha et al., 2007, 2011; Kathayat et al., 2017; Dutt et al., 2015)	Orbital-scale ISM variability; glacial–interglacial $\delta^{18}\text{O}$ amplitudes of several ‰	Wet-forest expansion and contraction; humid-refugium connectivity in Western Ghats and NE India; filtering of	Diversification slowdown at 3.0–2.5 Ma in Western Ghats ranoids (Cyriac et al., 2024); refugial patterns in S Western
		humid-adapted anurans and caecilians	Ghats (Joshi and Karanth, 2013); <i>Indirana semipalmata</i> demographic expansion (Kumar et al., 2020)
Speleothem $\delta^{18}\text{O}$, Heinrich and YD excursions [NE India] (Dutt et al., 2015)	Abrupt ISM weakening during H1–H3, YD; centennial excursions	Pulse-aridification stress on riparian and stream-breeding anurans (Nyctibatrachidae, Micrixalidae, Ranixalidae); expected demographic bottlenecks	No published test on Indian amphibians
Speleothem $\delta^{18}\text{O}$, 8.2 ka event [NE India] (Berkelhammer et al., 2012; Kathayat et al., 2018)	Abrupt monsoon weakening, ~150 yr duration	Short-term hydrological stress; localised range contraction in humid refugia	No published test on Indian herpetofauna
Speleothem and marine $\delta^{18}\text{O}$, 4.2 ka event [NW India, Arabian Sea] (Staubwasser et al., 2003; Dixit et al., 2014; Kathayat et al., 2018; Giesche et al., 2019)	~230 yr of stacked multi-decadal droughts; ~4‰ $\delta^{18}\text{O}$ shift in NW India; concurrent summer and winter monsoon weakening; reduced Indus discharge	Severe range contraction in dry-margin and Indus-plain herpetofauna; local extirpations possible	No direct test; post-Harappan hydrological transformation well established but herpetofaunal response undocumented
Arabian Sea foraminiferal proxies [NW Arabian Sea] (Gupta et al., 2003; Govil and Naidu, 2011; Saraswat et al., 2013; Giesche et al., 2019)	Monsoon-driven upwelling, runoff, SST and salinity; 32 kyr coverage; Heinrich and YD resolution; summer vs winter monsoon deconvolution	Modulation of estuarine and deltaic habitats used by <i>Gavialis gangeticus</i> , <i>Crocodylus palustris</i> and coastal squamates	Circumstantial: historical range contraction of extant crocodilians in the Indo-Gangetic plain is documented but has not been causally attributed to specific Quaternary hydroclimatic excursions
Karewa pollen and diatom [Kashmir Valley]; Toshmaidan peat [Kashmir]	Glacial–interglacial vegetation turnover in Kashmir; post-LGM stepwise warming	Elevational range shifts in Himalayan <i>Duttaphrynus</i> , <i>Scutiger</i> , <i>Amolops</i>	Late Pliocene (~3.2 Ma) Thakkhola-basin divergence in <i>Duttaphrynus</i>

(Dodia et al., 1984; Agrawal et al., 1989; Singh, 1963; Singh and Agrawal, 1976)	and forest re-expansion	and <i>Cyrtodactylus</i> ; niche tracking along T and humidity gradients	<i>himalayanus</i> linked explicitly to Himalayan geology and palaeoclimate (Hofmann et al., 2026); deep phylogeographic breaks in <i>Cyrtodactylus</i> at major river valleys (Agarwal et al., 2014); SDMs for Eastern Himalayan amphibians including <i>Scutiger sikimensis</i> (Subba et al., 2018)
Peninsular Indian lacustrine records [S India] (Sandeep et al., 2017)	Holocene centennial-scale ISM variability in S India; solar-forcing signatures	Fine-scale habitat availability for Western Ghats stream breeders; predicted demographic oscillations	Signature of post-contraction demographic expansion in <i>Indirana semipalmata</i> (Kumar et al., 2020) consistent with Quaternary wet-forest dynamics; not yet tied to specific lacustrine events
Historical-period speleothem records [N and NE India] (Sinha et al., 2015; Kathayat et al., 2022)	Annually resolved ISM variability over the past ~2000 years; multi-decadal droughts synchronous with documented famines	Anthropogenic-era habitat change superimposed on natural hydroclimatic variability; historical range contractions	No systematic historical-period herpetofaunal reconstruction for India; museum-collection-based approaches untested
Narmada fluvio-lacustrine sediments and YTT datum [Central India] (Acharyya and Basu, 1993; Patnaik et al., 2009)	Middle to Late Pleistocene fluvial dynamics; 74 ka tephra datum; mosaic habitats	Record of semi-aquatic reptiles (<i>Colossochelys</i> , <i>Geoclemys</i> , <i>Crocodylus palaeindicus</i>); tracking of habitat mosaics	Fossil chelonians and crocodiles documented (Rio and Mannion, 2024); amphibian record under-explored
Kurnool MCG cave deposits [Andhra Pradesh] (Patnaik et al., 2008)	Late Pleistocene to Holocene assemblage; impoverishment after LGM aridity	Loss of humid-forest herpetofauna; survival of arid-tolerant taxa (<i>Varanus</i> , <i>Python</i> , cf. <i>Duttaphrynus melanostictus</i>)	Direct fossil evidence: Late Pleistocene wooded-grassland herpetofauna replaced by modern arid-shrubland assemblage
Long-term tectono-climatic records [Himalaya, Arabian Sea, BoB] (Clift et al., 2008)	Coupling of Himalayan exhumation and Asian monsoon intensity over 23 Myr; Late Plio–Pleistocene intensification	Context for Neogene aridification and C4 expansion; habitat for open-country reptile clades	Reconciled with <i>Hemidactylus</i> and fan-throated lizard radiations in peninsular India (Lajmi and Karanth, 2020; Deepak and Karanth, 2018)
Fossil crocodilian record [Siwaliks and Narmada, India] (Rio and Mannion, 2024; Delfino and de Vos, 2010)	Late Miocene to Middle Pleistocene persistence of <i>Crocodylus palaeindicus</i> in India; Pliocene eastward dispersal of <i>Gavialis</i> from Indian origin	Biogeographic connectivity of South and Southeast Asian wetlands; response of large semi-aquatic reptiles to aridification	Direct fossil evidence; <i>C. palustris</i> appearance in Pleistocene consistent with faunal turnover

yielding warm-humid intervals during which humid-forest habitats expand. At millennial scale, the ISM weakened abruptly during Heinrich events H1–H3 and the Younger Dryas, captured as positive $\delta^{18}\text{O}$ excursions in the Mawmluh record (Dutt et al., 2015). At centennial to decadal scale, the 8.2 ka weakening and the 4.2 ka megadrought are resolved at annual precision in northeastern Indian speleothems (Berkelhammer et al., 2012; Kathayat et al., 2018), with the 4.2 ka event characterised as a ~ 230 -yr interval of stacked multi-decadal droughts (Kathayat et al., 2018).

3.2. Marine sedimentary archives: Arabian Sea and Bay of Bengal

Deep-sea sediment cores from the northern Arabian Sea and Bay of Bengal extend and complement the terrestrial speleothem network. These archives track monsoon-driven upwelling, primary productivity, runoff and sea-surface conditions. Gupta et al. (2003) pioneered the use of the planktic foraminifer *Globigerina bulloides* from ODP Site 723A as a proxy for Arabian Sea upwelling intensity, demonstrating that abrupt Holocene weakening events in the southwest monsoon coincided with cold phases in the North Atlantic and implying a hemispheric-scale atmospheric teleconnection. Saraswat et al. (2013) used planktic foraminiferal $\delta^{18}\text{O}$ and Mg/Ca from the eastern Arabian Sea to reconstruct last-glacial-to-Holocene sea-surface temperature and salinity, revealing that deglacial warming in the tropical Indian Ocean was paced by the interplay of regional monsoon and global teleconnections. Govil and Naidu (2011) quantified monsoon-precipitation variability of the last 32 kyr through surface hydrography of the western Bay of Bengal, linking reduced ISM rainfall to Heinrich events and the Younger Dryas. Staubwasser et al. (2003) documented a prominent positive *G. ruber* $\delta^{18}\text{O}$ excursion at 4.2 ka off the Indus delta, consistent with sharply reduced river discharge and coherent with the termination of urban Harappan occupation of the region.

A recent re-examination of the Indus delta record by Giesche et al. (2019) used three planktic foraminiferal species with distinct ecological niches to deconvolve summer and winter monsoon signals across the 4.2 ka event. This record shows that the summer monsoon weakening was accompanied, and arguably amplified, by a concurrent decrease in winter precipitation by ~ 4.1 ka. The implication is that year-round water availability collapsed across the plains of northwest South Asia at the transition from urban to post-urban Indus phase. Dixit et al. (2014) corroborated the terrestrial signal from continental lake carbonate $\delta^{18}\text{O}$ in northwest India, where the 4.2 ka event is expressed as an abrupt $\sim 4\text{‰}$ positive shift; subsequent gypsum hydration water $\delta^{18}\text{O}$ and δD from palaeolake Karsandi constrain earlier mid-Holocene intensification at 5.1–4.4 ka (Dixit et al., 2018). Together, these marine and fluvial isotope archives define a mid-to late-Holocene trajectory of intensification followed by abrupt weakening and sustained aridification, with Heinrich-scale monsoon excursions superimposed on orbitally forced glacial–interglacial variability.

Coherence of 4.2 ka proxy signals across speleothem (Kathayat et al., 2018), foraminiferal (Staubwasser et al., 2003; Giesche et al., 2019), lacustrine (Dixit et al., 2014) and palaeolake (Dixit et al., 2018) archives, combined with the archaeologically documented deurbanisation of the Indus Civilisation, constitutes an important interdisciplinary context for Quaternary climate–society interaction on the Subcontinent. The Indus plain hosts extant populations of *Gavialis gangeticus* (critically endangered), *Crocodylus palustris*, and several monitor lizards and colubrid snakes. Whether any of these populations responded to 4.2 ka hydroclimatic change remains, to our knowledge, untested in the published herpetological literature; future palaeoenvironmental sampling at Indus-plain and adjacent peninsular sites could address this gap.

On longer timescales, Clift et al. (2008) used weathering records from the South China Sea, Bay of Bengal and Arabian Sea to demonstrate a dynamic coupling between Himalayan exhumation rate and Asian monsoon intensity over the past 23 Myr, with renewed intensification in the Late Pliocene–Pleistocene. This long-term context frames the

tectono-climatic backdrop against which Indian Quaternary biotic evolution occurred.

3.3. Lacustrine and pollen archives

The Karewa Group of the Kashmir Valley preserves ~ 4 Myr of lacustrine, fluvial and loess deposition. Magnetostratigraphic and luminescence chronologies place Lower Karewa sedimentation in the Plio–Pleistocene and Upper Karewa deposition in the Middle to Late Pleistocene (Agrawal et al., 1989). Pollen and diatom assemblages document repeated transitions between warm-humid, forested interglacial intervals and cold, open, conifer-dominated glacial intervals (Dodia et al., 1984; Agrawal et al., 1989). Post-LGM vegetation reconstructed from high-altitude peat bogs (Toshmaidan) shows stepwise postglacial warming, a Holocene climatic optimum, and late Holocene re-expansion of *Pinus wallichiana* and *Betula* (Singh, 1963; Singh and Agrawal, 1976). In peninsular southern India, Sandeep et al. (2017) integrated multi-proxy lake sediment records through the Holocene, documenting centennial-scale ISM variability and solar forcing signatures comparable to those in the marine record. Ganga-plain pollen archives (Sharma et al., 2004) constrain late-glacial to Holocene vegetation turnover that frames faunal response.

3.4. Fluvial and alluvial archives: Siwalik and Narmada

The Siwalik Group records Neogene to Early Pleistocene foreland-basin sedimentation along the Himalayan front and preserves vertebrate-rich fluvial deposits (Patnaik, 2003). Upper Siwalik micro-vertebrate assemblages from Jammu and adjacent regions indicate broadly well-watered, wooded-grassland palaeoenvironments transitioning to drier, more seasonal regimes through the Plio–Pleistocene, consistent with Late Neogene monsoon intensification (Patnaik, 2003; see also Clift et al., 2008, for regional tectono-climatic context). The Central Narmada Valley exposes up to ~ 130 m of Pleistocene fluvio-lacustrine sediments. Palaeomagnetic and tephra-based chronologies place the exposed succession within at least the last ~ 700 kyr, with the Youngest Toba Tuff (YTT) providing a ~ 74 ka chronostratigraphic marker in the middle of the sequence (Acharyya and Basu, 1993; Patnaik et al., 2009). Lens-shaped pond-bank beds at Bhedaghat and Devakachar preserve stream, pond, wooded-grassland and semi-arid microhabitats in close stratigraphic association, constituting one of the best-resolved Indian terrestrial archives of Middle to Late Pleistocene environmental mosaics (Patnaik et al., 2009).

3.5. The Toba supereruption and its chronostratigraphic marker

The ~ 74 ka Toba eruption deposited the YTT across peninsular India as an isochronous marker horizon in many river valleys (Petruglia et al., 2007; Jones, 2012). Interdisciplinary fieldwork in the Jurreru and Middle Son valleys shows that a mosaic of environments persisted across the YTT, that local drying was already under way before the eruption, and that Middle Palaeolithic archaeology spans the event (Petruglia et al., 2007). Global syntheses now conclude that climatic and ecological impacts of Toba may have been overestimated (Yost et al., 2018). For Quaternary science, however, the YTT remains the single most powerful chronostratigraphic datum in peninsular Indian Pleistocene stratigraphy, and is routinely used to correlate fluvial, alluvial and cave-deposit sequences across the Subcontinent.

3.6. Historical-period archives: the past two millennia

The past two millennia occupy a special place in Indian Quaternary research because they couple precisely dated proxy records to documentary archives of famine, monsoon failure and societal change. Sinha et al. (2015) reconstructed ISM rainfall variability over the last ~ 2000 years from a composite Sahiya cave (north India) speleothem $\delta^{18}\text{O}$

record, showing that recent multi-decadal declines in summer rainfall fall within the range of natural variability. Kathayat et al. (2022) presented an annually resolved Mawmluh cave $\delta^{18}\text{O}$ record in which protracted multi-decadal droughts of the past millennium are synchronous with historically documented famines, mass-mortality events and geopolitical transitions in the Indian subcontinent. These historical archives close the resolution gap between Pleistocene proxy work and the instrumental era, and provide direct baselines for the Anthropocene trajectories discussed in Section 7. From the perspective of herpetofaunal biogeography, they also define a period during which intense land-use intensification, deforestation and hydraulic modification began to overlay natural climatic signals. This palimpsest will need to be disentangled in future research on historical herpetofaunal range contractions.

4. The Quaternary herpetofaunal record: direct archival evidence

Fossil and subfossil amphibian and reptile remains form a biogenic assemblage complementary to pollen, foraminiferal and mammalian archives, recording Quaternary environmental change directly through the occurrence and morphology of taxa with known habitat requirements. This section reviews the three principal Indian Quaternary sites at which herpetofaunal remains have been described, with attention to the taphonomic limits that govern inference. These three localities Siwalik, Narmada and Kurnool are treated here because they are effectively the only Indian Quaternary sites with described, stratigraphically constrained herpetofaunal material. Other Quaternary vertebrate localities of the Subcontinent, including sites in Gujarat and Bihar and the predominantly Upper Pleistocene cave and open-air deposits of Sri Lanka, have yielded little or no reported amphibian and reptile material and are noted where relevant (Chauhan, 2008) rather than reviewed separately a paucity that itself reflects the taphonomic and collection biases discussed in Section 4.4.

4.1. Siwalik Group

Herpetofaunal remains in the Siwalik Group are best documented for crocodylids. *Crocodylus palaeindicus* is restricted to the late Miocene to early Middle Pleistocene of India, with the extant mugger *C. palustris* appearing unequivocally in the fossil record only from the Pleistocene (Rio and Mannion, 2024). The gharial *Gavialis* first appears in the Subcontinent in the Pliocene and dispersed east along the Siva–Malayan route during the Quaternary, with Late Pleistocene records of *G. bengawanicus* on Java (Delfino and de Vos, 2010). Amphibian and squamate remains from Siwalik sediments are less systematically treated; their underrepresentation is a taphonomic rather than a biological signal (see Section 4.4).

4.2. Narmada valley pleistocene succession

The Narmada Valley Pleistocene succession (Section 3.4) yields the most direct Indian terrestrial herpetofossil record that can be coupled to palaeoclimatic inference. Reptilian elements include abundant chelonians, specifically *Colossochelys* and the freshwater *Geoclemys*, together with the extinct crocodyloid *Crocodylus palaeindicus* (Rio and Mannion, 2024; Patnaik et al., 2009). The Narmada record is unique on the Subcontinent in combining a long chronology with YTT chronostratigraphy and preservation of semi-aquatic reptiles, though systematic screen-washing for amphibian microfossils has not yet been reported.

4.3. Kurnool caves

Late Pleistocene to Early Holocene deposits at Muchchatla Chintamanu Gavi (MCG), Kurnool district, spanning ~20 kyr BP, preserve a herpetofaunal component of *Python* sp., indeterminate serpentes,

Varanus sp., and an anuran referred to cf. *Duttaphrynus melanostictus*, alongside small mammals and birds (Patnaik et al., 2008). The Late Pleistocene fauna implies wooded grassland with gallery forest and a well-watered landscape. Today the surrounding area is arid shrubland, and only a fraction of the Late Pleistocene vertebrate fauna persists locally. The Late Pleistocene–Holocene faunal impoverishment has been attributed, at least partly, to severe aridity at the LGM (Patnaik et al., 2008). Elsewhere in peninsular India, Neogene herpetofaunal evidence older than the Middle Pleistocene is largely absent outside Gujarat and Bihar, and Sri Lankan cave and open-air deposits are predominantly Upper Pleistocene (Chauhan, 2008).

4.4. Taphonomy: interpreting absence

Across all Indian Quaternary sites, the herpetofaunal record is filtered by three biases. Preservation bias means that frog and lizard bones are small, porous and fragile, and fracture or dissolve under the depositional conditions prevailing in most Indian Quaternary fluvial systems. Where microvertebrate screen-washing has been applied (Narmada lens beds, MCG), herpetofaunal taxa emerge; where it has not, they are effectively invisible. Collection bias means that historical excavations at Narmada, Kurnool and the Siwaliks privileged large mammalian and hominin material. Chronological bias means that few Indian herpetofaunal assemblages have direct absolute dating, limiting correlation to precisely dated monsoon events. Accurate presence/absence inference for Indian Pleistocene herpetofauna therefore requires explicit taphonomic evaluation, small-vertebrate-targeted sampling, matrix sieving, and direct U-series or amino-acid racemisation dating of bone (Turvey et al., 2021). The principal opportunity is the Narmada sequence, whose YTT datum and long chronology could in principle anchor a Middle to Late Pleistocene Indian herpetofaunal biostratigraphy.

5. Phylogeographic signatures: the biotic test

5.1. Pre-Quaternary baseline

Any Quaternary interpretation of Indian herpetofauna must acknowledge a deep-time baseline. Most major Indian herpetofaunal clades originated well before the Quaternary. Gondwanan relict lineages entered the Asian biota during the Palaeogene Indian-plate drift and post-collision phase (Bossuyt and Milinkovitch, 2001; Biju and Bossuyt, 2003; Van Bocxlaer et al., 2006, 2009); rhacophorid and bufonid radiations date to the Eocene–Miocene (Li et al., 2013); peninsular *Hemidactylus*, *Cyrtodactylus* (*Geckoella*) and fan-throated lizards diversified under Neogene cooling and aridification driven by Himalayan uplift and monsoon intensification (Agarwal and Karanth, 2015; Deepak and Karanth, 2018; Lajmi and Karanth, 2020; Clift et al., 2008). The Quaternary signal in Indian herpetofauna is therefore not one of clade origination but of intra-clade sorting, range adjustment and demographic response: a shallower signal that must be sought with methods matched to that scale.

5.2. Western Ghats–Sri Lanka: Quaternary refugial dynamics

The Western Ghats and Sri Lanka constitute the Subcontinent's best-sampled region for proxy-driven tests of Quaternary impact on herpetofauna. Current taxonomy recognises ~252 amphibian species in the Western Ghats, three endemic families (Micrixalidae, Nasikabatrachidae, Ranixalidae) and 16 endemic genera (Dahanukar and Molur, 2020); Sri Lanka adds a major endemic radiation of the shrub-frog genus *Pseudophilautus* (Meegaskumbura et al., 2002, 2019). Endemism is interpreted in a refugial framework: the southern Western Ghats buffered humid-adapted lineages through repeated Late Cenozoic climatic oscillations (Joshi and Karanth, 2013; Robin et al., 2010).

Vijayakumar et al. (2016) used a dated phylogeny of *Raorchestes*

bush frogs to evaluate Quaternary glaciations, ecological gradients and geological barriers simultaneously, concluding that topography, latitudinal seasonality gradients and glacial–interglacial fluctuations jointly structured diversification. Broader analyses of ranoid frogs qualify any “species pump” expectation: most clades show constant or declining diversification rates through time, with a slowdown at ~3.0–2.5 Ma during Pleistocene glaciation cycles rather than a Pleistocene acceleration (Cyriac et al., 2024). The mechanistic chain implied is: orbital-scale ISM weakening, followed by humid-forest contraction, followed by population fragmentation without new species origination. This orbital-scale inference is directly consistent with the glacial–interglacial amplitudes documented in Indian speleothems (Kathayat et al., 2017; Dutt et al., 2015) and in Karewa pollen (Agrawal et al., 1989). At shallower depth, Kumar et al. (2020) documented in *Indirana semipalmata* a signature of high haplotype diversity and low nucleotide diversity consistent with post-contraction demographic expansion, together with strong genetic structure across the Shencottah Gap. This constitutes a direct empirical link between palaeoclimatic contraction and contemporary genetic structure in a Western Ghats endemic.

The Sri Lankan *Pseudophilautus* radiation provides an independent test. Meegaskumbura et al. (2019) dated colonisation to a cool, dry Eocene–Oligocene (Oi-1) land-bridge phase, with diversification through the Miocene and early Pliocene and a minor Pleistocene burst. These dynamics are consistent with Quaternary low-sea-level land-bridge connections across the Palk Strait repeatedly facilitating biotic exchange between peninsular India and Sri Lanka (Meegaskumbura et al., 2002, 2019).

5.3. Himalayan and Northeast Indian herpetofauna

Hofmann et al. (2026) used ddRAD genomic sequencing and palaeoclimatic niche modelling to reconstruct the diversification of the Himalayan toad *Duttaphrynus himalayanus*, a widespread montane amphibian. Two deeply diverged phylogeographic lineages are identified east and west of the Thakkhola basin (upper Kali Gandaki River), with divergence dated to the Late Pliocene (~3.2 Ma). This result provides perhaps the most direct published integration of geological, palaeoclimatic and phylogeographic evidence for any Indian Subcontinent amphibian, and the authors invoke palaeolake corridors formed during the Late Quaternary as potential connectivity windows. *Cyrtodactylus* geckos show a consistent pattern of in situ Himalayan diversification with deep phylogenetic breaks at major river valleys (Agarwal et al., 2014). In Northeast India, the endemic caecilian family Chikilidae is nested within an African Herpelidae clade, implying an Early Cretaceous (~140 Ma) divergence and demonstrating that the region retains deep, under-documented lineages (Kamei et al., 2012). Elevational distributions of Eastern Himalayan amphibians depend strongly on temperature and humidity, making these communities particularly vulnerable to warming and upslope habitat loss (Subba et al., 2018).

5.4. Reptile signatures of aridification

For reptiles, the dominant macroevolutionary signal is Neogene aridification rather than Quaternary glaciation. C4 grassland expansion across the Late Miocene–Pliocene, documented by Siwalik soil-carbonate and tooth-enamel isotopes and by regional monsoon intensification (Clift et al., 2008), opened habitat for open-country lineages. Fan-throated lizards diversified from ~39 to 32 Ma with habitat-specific radiation accelerating under Neogene seasonality (Deepak and Karanth, 2018). *Hemidactylus* geckos show an early burst at Eocene–Oligocene cooling and slowed diversification through the Neogene (Lajmi and Karanth, 2020). Within these clades the Quaternary contributed range restriction and population structuring rather than new speciation. Arid-adapted lineages likely expanded during glacial intervals and contracted during humid interglacials, a testable prediction against the

reconstructed ISM amplitude (Kathayat et al., 2017).

Several reptile clades provide independent, if coarser-grained, tests of these expectations. The ground-dwelling *Geckoella* radiation (nested within *Cyrtodactylus*) diversified across peninsular India and Sri Lanka in step with Neogene aridification and the breakup of once-continuous wet forest into isolated massifs (Agarwal and Karanth, 2015), a spatial structuring that Quaternary climatic oscillations are more likely to have reinforced than initiated. In the Himalaya, *Cyrtodactylus* shows deep, geographically partitioned lineages separated by major river valleys (Agarwal et al., 2014) the same physiographic barriers implicated in the Late Pliocene divergence of *Duttaphrynus himalayanus* (Hofmann et al., 2026) so that gecko and toad breaks coincide across shared barriers and jointly implicate Late Cenozoic drainage and climate in structuring montane herpetofaunal diversity. Among semi-aquatic reptiles, the persistence of *Crocodylus palaeindicus* from the late Miocene into the Middle Pleistocene, the first unequivocal Pleistocene appearance of the extant mugger *C. palustris* (Rio and Mannion, 2024), and the Pliocene eastward dispersal of *Gavialis* along the Siva–Malayan route (Delfino and de Vos, 2010) together track the availability and connectivity of large perennial wetlands under a fluctuating monsoon. For reptiles, therefore, the Quaternary signal is expressed less as speciation than as range adjustment, lineage sorting and local extirpation superimposed on radiations of Neogene age a pattern that mirrors, at a different tempo, the refugial dynamics documented for wet-zone anurans.

5.5. Caecilians: deep roots, opaque Quaternary histories

India hosts 41 caecilian species across three of its four biodiversity hotspots (Dinesh et al., 2023). With limbless, fossorial ecology and restricted dispersal capacity, caecilians retain deep vicariance signatures. In the Western Ghats, Gower et al. (2016) found Eastern–Western Ghats *Gegeneophis* divergence too ancient to be explained by Miocene wet-zone contraction or Pleistocene climatic fluctuations, reinforcing that Pleistocene climate change acted on Indian caecilians at population and intra-clade levels rather than at the species-origination level.

6. Multi-proxy synthesis: proxy, mechanism and response framework

Fig. 2 summarises the proxy, mechanism and response framework schematically, and Table 1 operationalises it. For each class of proxy record, Table 1 identifies (a) the proxy and its geographic scope on the Subcontinent, (b) the quantitative signal and its timing, (c) the causal mechanism by which the signal is expected to affect amphibians and reptiles, and (d) the extent to which Indian herpetofaunal observations empirically corroborate, contradict or do not yet test the expected response. Rows are colour-coded according to empirical support: green denotes direct empirical evidence, amber denotes partial or circumstantial support, and red denotes rows with no direct published test; these define the review's empirical frontier.

6.1. What the framework reveals

Three patterns emerge. Where the proxy signal is orbital-scale, biological evidence is recoverable: Plio–Pleistocene refugial dynamics in the Western Ghats are empirically tractable (Vijayakumar et al., 2016; Cyriac et al., 2024) because dated phylogenies resolve events on that timescale, and the Himalayan *Duttaphrynus himalayanus* divergence is another orbital-scale case (Hofmann et al., 2026). Where the proxy signal is millennial, centennial or historical (Heinrich events, 8.2 ka, 4.2 ka, last millennium droughts), biological evidence in Indian herpetofauna is essentially empty despite the high precision of the archival record (red rows in Table 1). This is the single most striking mismatch. Where the proxy signal is high-magnitude and localised (Kurnool LGM aridity), direct fossil evidence shows community-level impoverishment (Patnaik et al., 2008), but the dominant responsible mechanism,

whether thermal stress, hydrological drought, or vegetation turnover, has not been disentangled.

6.2. Scale mismatch

A systematic scale mismatch underlies the framework. Indian herpetofaunal phylogeography has concentrated on Miocene–Pliocene events, while the best-resolved palaeoclimate records are Holocene and historical. Bridging them is a methodological challenge. Coalescent demographic estimators (Bayesian skyline, Stairway plots, extended Bayesian skyline) can in principle detect bottlenecks at 8.2 ka or 4.2 ka for species with short generation times. High-throughput genomic data are required for this resolution, and have so far been applied to very few Indian taxa; Kumar et al. (2020) and Hofmann et al. (2026) are among the few examples. Until such analyses are run on multiple co-distributed species, the millennial- and historical-scale rows of Table 1 will remain empty.

6.3. Geographic asymmetries

Dense proxy records cluster in northeastern India (Mawmluh, Sahiya, Wah Shikhar), northern India (Bitto, Tityana, Sainji) and the Kashmir–Karewa system, with peninsular contributions from Dandak, Jhumar and the Munagamanu–Valmiki caves, complemented by Arabian Sea sediment cores (Gupta et al., 2003; Govil and Naidu, 2011; Saraswat et al., 2013) and peninsular lakes (Sandeep et al., 2017) (Fig. 1). Dense herpetological phylogeography is concentrated in the Western Ghats and Sri Lanka. The geographic overlap between dense proxy and dense herpetofaunal sampling is therefore imperfect. The Himalayan arc, where Hofmann et al. (2026) provides a pioneering palaeoclimate–phylogenomic integration for *D. himalayanus*, along with Northeast India, the Indo-Gangetic plain, the Eastern Ghats and the Thar dry zone, are all underrepresented in one or both literatures. These regions define where targeted new work would deliver the greatest marginal gain.

6.4. Uncertainties, alternative explanations and the status of the framework

The links compiled in Table 1 are of unequal evidential strength, and most are best regarded as hypotheses to be tested rather than as demonstrated mechanisms. Three sources of uncertainty deserve emphasis. First, proxy uncertainty: speleothem $\delta^{18}\text{O}$ integrates moisture-source, amount and temperature effects that are not always separable; marine foraminiferal proxies are averaged by bioturbation and by seasonal and depth habitat preferences; and pollen spectra are filtered by differential production, transport and preservation. A single monsoon signal is therefore reconstructed with differing amplitude and timing by different archives, which do not always agree. Second, chronological uncertainty: age models for the herpetofaunal record are typically much coarser than those for the physical proxies, so an apparent temporal coincidence between a climatic excursion and a biotic change may reflect dating imprecision rather than a causal connection. Where this review states that a proxy signal aligns with a diversification slowdown or a demographic event, that alignment is temporal and correlational; competing explanations—tectonic and drainage rearrangement, biotic interactions, sampling and preservation bias, or stochastic lineage sorting—are seldom excluded by the data at hand. Third, spatial heterogeneity: proxy archives and phylogeographic sampling are unevenly distributed and only partly overlapping (Section 6.3), so Subcontinent-wide inferences are extrapolated from a geographically biased base. The framework is accordingly best read as a structured, hypothesis-generating device: its value lies in making the proxy–biota links explicit and falsifiable, and in exposing where evidence is absent—the red rows of Table 1—rather than in asserting established causation.

7. From quaternary baselines to Anthropocene trajectories

The Quaternary palaeoenvironmental record provides a quantitative baseline against which ongoing change can be calibrated. Indian herpetofauna persisted through repeated Pleistocene glacial–interglacial cycles, abrupt millennial monsoon excursions and the Toba super-eruption. That persistence was enabled by three features that no longer fully apply: continuous humid refugia in the southern Western Ghats and Sri Lanka, repeated land-bridge connections across the Palk Strait during glacial low stands, and terrestrial landscapes not fragmented by agriculture, roads or urban expansion. The historical speleothem records (Sinha et al., 2015; Kathayat et al., 2022) show that natural multi-decadal monsoon failures in India were severe enough to be contemporaneous with documented famines and geopolitical transitions. The Anthropocene signature is qualitatively different because it combines such natural variability with systematic warming, land-use change and habitat fragmentation. Current assessments indicate that climate change accounts for ~39% and habitat loss ~37% of recent deteriorations in amphibian Red List status globally, with the Western Ghats–Sri Lanka region among the most severely affected (Luedtke et al., 2023). Ecological niche modelling for Western Ghats squamates projects declines in climatic suitability for the majority of endemic threatened lizards and snakes under both RCP4.5 and RCP8.5, and novel protected-area expansion will be required to maintain representation (Srinivasulu et al., 2021). Himalayan amphibians face compounded risks: direct modelling of limited migration shows that even conservative dispersal constraints sharply reduce projected suitable areas (Subba et al., 2018).

Three features distinguish the Anthropocene challenge from Quaternary analogues. Rate is one: current warming and monsoon-variability change rates exceed the average Quaternary glacial–interglacial transition by roughly an order of magnitude (Diffenbaugh and Field, 2013), compressing the time window—and raising the spatial velocity of climate—over which populations must track suitable conditions (Loarie et al., 2009). Combinatorial novelty is another: past climatic stress was not paired with pathogens such as *Batrachochytrium dendrobatidis*, with agro-industrial habitat fragmentation, or with road and canal networks that interrupt land-bridge and riparian connectivity. Refugial integrity is a third: Quaternary refugia were climatic rather than anthropogenic, and their integrity was maintained across glacial cycles; today, those refugia are themselves shrinking under land-use pressure.

The framework developed here is designed to inform future projections for the Indian Subcontinent in three practical ways. First, the Quaternary proxy network establishes the amplitude and tempo of natural ISM variability against which anthropogenic perturbations can be benchmarked; projected 21st-century Indian monsoon change must be interpreted relative to this natural envelope rather than against instrumental records alone. Second, Quaternary refugia identified through phylogeographic and palaeoenvironmental congruence (Western Ghats wet-zone, Eastern Himalayan valleys, peninsular riparian corridors) are candidate climate-change refugia for the Anthropocene and should be prioritised in protected-area design. Third, proxy-calibrated hindcast species distribution models, of the kind outlined in Section 8, provide a methodologically defensible bridge between Quaternary-to-Recent analogues and future projections, reducing reliance on modern-only calibration under non-analogue climate conditions.

This Quaternary-to-Recent perspective on Indian herpetofauna contributes directly to two of the United Nations' Sustainable Development Goals. SDG 13 (Climate Action) requires credible baselines for ongoing change; the Indian Quaternary proxy network, as read through the biogenic assemblages evaluated here, supplies such baselines at time-scales the instrumental record cannot reach. SDG 15 (Life on Land) targets the conservation of terrestrial ecosystems and the species they support; Indian amphibians and reptiles are both sentinels of and

participants in these ecosystems, and their Quaternary trajectories inform present-day priority-setting. A herpetofaunal perspective on Indian Quaternary environmental change, delivered alongside conventional palaeoenvironmental analyses, adds a biotic dimension to the climate-society narrative that this special issue articulates.

8. Knowledge gaps and research priorities

Five priorities emerge from the framework in Table 1.

- (i) Microvertebrate-targeted fossil recovery from Narmada, Kurnool and Karewa sequences, using systematic screen-washing and modern absolute-dating methods such as uranium-series, amino-acid racemisation, and direct bone apatite dating (Turvey et al., 2021). The Narmada succession, with its YTT datum, is the single most promising site for establishing a Middle to Late Pleistocene Indian herpetofaunal biostratigraphy.
- (ii) Proxy-calibrated population genomics. Co-distributed Indian amphibians and reptiles with markers suitable for demographic inference, including *Indirana*, *Nyctibatrachus*, *Raorchestes*, *Uperodon*, *Cnemaspis*, *Hemidactylus* and *Duttaphrynus*, should be sampled across their ranges. Coalescent bottleneck and expansion tests can then be formally compared to Heinrich, YD, 8.2 ka and 4.2 ka intervals reconstructed from Indian speleothems (Dutt et al., 2015; Kathayat et al., 2018). Pioneering single-species examples include Kumar et al. (2020) on *Indirana semipalmata* and Hofmann et al. (2026) on *Duttaphrynus himalayanus*.
- (iii) Proxy-calibrated species distribution modelling. Hindcast SDMs for Indian herpetofauna to the LGM, mid-Holocene and 4.2 ka event, informed by speleothem- and marine-proxy-constrained rainfall minima (Saraswat et al., 2013; Govil and Naidu, 2011; Kathayat et al., 2018; Giesche et al., 2019), would test whether observed phylogeographic structure is consistent with reconstructed habitat availability. Subba et al. (2018) and Hofmann et al. (2026) provide Himalayan templates.
- (iv) Geographic infill. The under-sampled regions identified in Section 6.3 (the Himalayan arc, Northeast India, the Eastern Ghats, the Indo-Gangetic plain and the Thar dry zone) require parallel deepening of palaeoclimate and herpetofaunal sampling. Low-dispersal taxa most likely to retain Quaternary signals, such as fossorial reptiles, small ground skinks, stream-breeding microhylids and caecilians, should be prioritised.
- (v) Historical-period reconstruction. Museum collections, colonial-era faunistic records and historical ecological inventories should be systematically compiled to build a ~200-yr baseline

against the Sinha et al. (2015) and Kathayat et al. (2022) historical speleothem records. Integrated data products, such as community-level Indian Quaternary herpetofaunal databases linked to dated palaeoclimate time series in the manner of the SISAL speleothem database, would convert a dispersed literature into a reusable, hypothesis-testing resource.

9. Conclusions

The Indian Subcontinent preserves one of the most complete networks of Quaternary palaeoenvironmental archives on Earth. Speleothem, marine, lacustrine, fluvial and cave-deposit records together resolve Indian Summer Monsoon variability from orbital to annual timescales, and document abrupt Holocene and historical excursions at a precision unmatched in most tropical settings. Read against this proxy backbone, the Subcontinent's endemic herpetofauna forms a partly exploited biotic archive. The orbital-scale imprint of Quaternary climate on Indian amphibians and reptiles is tractable and partly documented: Plio-Pleistocene diversification slowdowns, Western Ghats and Sri Lanka refugial dynamics, peninsular cave-fauna impoverishment after the LGM, and deep Himalayan phylogeographic breaks in *Cyrtodactylus* and *Duttaphrynus himalayanus* at major river valleys. The millennial, centennial and historical imprint remains largely untested despite the high precision of the relevant proxy records. Closing this gap requires coordinated microvertebrate taphonomy, proxy-calibrated population genomics and SDM hindcasting across the Subcontinent. Because Indian amphibians and reptiles are simultaneously sentinels of environmental change and among the most threatened vertebrate groups in the region today, this agenda is not only scientifically tractable but also conservation-critical. The framework presented here is intended to be useable by Quaternary geologists, climatologists, archaeologists and biodiversity researchers as they plan future interdisciplinary work on Indian Quaternary environmental change.

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Box 1

Key messages for interdisciplinary audiences

The framework developed in this review yields six take-home messages intended to serve geologists, climatologists, archaeologists, environmentalists, educators and early-career researchers.

- (1) **Indian Quaternary proxy archives are globally outstanding.** Speleothem, marine, lacustrine and fluvial records resolve ISM variability from orbital to annual timescales across the Subcontinent.
- (2) **Herpetofauna are an underused biogenic archive.** Amphibians and reptiles of the Western Ghats, Sri Lanka, the Himalaya and Northeast India retain phylogeographic signatures of Plio-Pleistocene climate, but are rarely read alongside proxy records.
- (3) **Orbital-scale signals are tractable; millennial signals are not.** The review identifies a systematic mismatch between the resolution of Quaternary proxy archives and the resolution of published biotic analyses.
- (4) **The 4.2 ka event has interdisciplinary pay-off.** Speleothem, marine and palaeolake archives converge on an abrupt monsoon failure coincident with the Indus Civilisation transition; the herpetofaunal response remains untested.
- (5) **Quaternary refugia matter for Anthropocene conservation.** Refugia identified by proxy-biogenic congruence are candidate climate-change refugia and should inform protected-area planning.
- (6) **This agenda supports UN SDGs 13 and 15.** A biogenic perspective on Indian Quaternary environmental change adds a life-on-Earth dimension to climate-action and biodiversity-conservation decision-making.

Data availability

All data and/or code is contained within the submission.

References

- Acharyya, S.K., Basu, P.K., 1993. Toba ash on the Indian subcontinent and its implications for correlation of late Pleistocene alluvium. *Quat. Res.* 40 (1), 10–19. <https://doi.org/10.1006/qres.1993.1051>.
- Agarwal, I., Bauer, A.M., Jackman, T.R., Karanth, K.P., 2014. Insights into Himalayan biogeography from geckos: a molecular phylogeny of *Cyrtodactylus* (Squamata: gekkonidae). *Mol. Phylogenet. Evol.* 80, 145–155. <https://doi.org/10.1016/j.ympev.2014.07.018>.
- Agarwal, I., Karanth, K.P., 2015. A phylogeny of the only ground-dwelling radiation of *Cyrtodactylus* (Squamata, Gekkonidae): diversification of *Geckoella* across peninsular India and Sri Lanka. *Mol. Phylogenet. Evol.* 82, 193–199. <https://doi.org/10.1016/j.ympev.2014.09.016>.
- Agrawal, D., Dodia, R., Kotlia, B., Razdan, H., Sahni, A., 1989. The Plio-Pleistocene geologic and climatic record of the Kashmir valley, India: a review and new data. *Palaeogeogr. Palaeoclimatol. Palaeoecol.* 73 (3–4), 267–286. [https://doi.org/10.1016/0031-0182\(89\)90008-4](https://doi.org/10.1016/0031-0182(89)90008-4).
- Badam, G.L., 1979. Pleistocene fauna of India. Deccan College Postgraduate and Research Institute.
- Banerji, U.S., Arulbalaji, P., Padmalal, D., 2020. Holocene climate variability and Indian summer Monsoon: an overview. *Holocene* 30 (5), 744–773. <https://doi.org/10.1177/0959683619895577>.
- Behrensmeyer, A.K., 1978. Taphonomic and ecologic information from bone weathering. *Paleobiology* 4 (2), 150–162. <https://doi.org/10.1017/S0094837300005820>.
- Berkelhammer, M., Sinha, A., Stott, L., Cheng, H., Pausata, S.R., Yoshimura, K., 2012. An abrupt shift in the Indian Monsoon 4000 years ago, 75–88. <https://doi.org/10.1029/2012GM001207>.
- Biju, S.D., Bossuyt, F., 2003. New frog family from India reveals an ancient biogeographical link with the Seychelles. *Nature* 425 (6959), 711–714. <https://doi.org/10.1038/nature02019>.
- Bossuyt, F., Milinkovitch, M.C., 2001. Amphibians as indicators of early tertiary "Out-of-India" dispersal of vertebrates. *Science* 292 (5514), 93–95. <https://doi.org/10.1126/science.1058875>.
- Chauhan, P.R., 2008. Large mammal fossil occurrences and associated archaeological evidence in Pleistocene contexts of peninsular India and Sri Lanka. *Quat. Int.* 192 (1), 20–42. <https://doi.org/10.1016/j.quaint.2007.06.034>.
- Clift, P.D., Hodges, K.V., Heslop, D., Hannigan, R., Van Long, H., Calves, G., 2008. Correlation of Himalayan exhumation rates and Asian monsoon intensity. *Nat. Geosci.* 1 (12), 875–880. <https://doi.org/10.1038/ngeo351>.
- Cox, N., Young, B.E., Bowles, P., Fernandez, M., Marin, J., Rapacciuolo, G., et al., 2022. A global reptile assessment highlights shared conservation needs of tetrapods. *Nature* 605 (7909), 285–290. <https://doi.org/10.1038/s41586-022-04664-7>.
- Cyriac, V.P., Mohan, A.V., Dinesh, K.P., Torsekar, V., Jayarajan, A., Swamy, P., Vijayakumar, S.P., Shanker, K., 2024. Diversifying in the mountains: spatiotemporal diversification of frogs in the Western Ghats biodiversity hotspot. *Evolution* 78 (4), 701–715. <https://doi.org/10.1093/evolut/qpae006>.
- Dahanukar, N., Molur, S., 2020. JoTT checklist of amphibians of the Western Ghats (v1.0). J. Threat. Taxa. <https://threatenedtaxa.org/index.php/JoTT/checklists/amphibians/westernghats>.
- Deepak, V., Karanth, K.P., 2018. Aridification driven diversification of fan-throated lizards from the Indian subcontinent. *Mol. Phylogenet. Evol.* 120, 53–62. <https://doi.org/10.1016/j.ympev.2017.11.016>.
- Delfino, M., de Vos, J., 2010. A revision of the Dubois crocodylians, *Gavialis bengawanicus* and *Crocodylus ossifragus*, from the Pleistocene *Homo erectus* beds of Java. *J. Vertebr. Paleontol.* 30 (2), 427–441. <https://doi.org/10.1080/02724631003617910>.
- Diffenbaugh, N.S., Field, C.B., 2013. Changes in ecologically critical terrestrial climate conditions. *Science* 341 (6145), 486–492. <https://doi.org/10.1126/science.1237123>.
- Dinesh, K.P., Deuti, K., Saikia, B., 2023. Fauna of India checklist: amphibia. *Zool. Surv. India*. <https://doi.org/10.26515/Fauna/1/2023/Chordata:Amphibia>.
- Dinesh, K.P., Radhakrishnan, C., Gururaja, K.V., Deuti, K., Bhatta, G.K., 2009. An annotated checklist of Amphibia of India with some insights into the patterns of species discoveries, distribution and endemism. *Record Zool. Surv. India Occas. Pap.* 302, 1–153.
- Dixit, Y., Hodell, D.A., Giesche, A., Tandon, S.K., Gázquez, F., Saini, H.S., Skinner, L.C., Mujtaba, S.A.I., Pawar, V., Singh, R.N., Petrie, C.A., 2018. Intensified summer monsoon and the urbanization of Indus Civilization in northwest India. *Sci. Rep.* 8 (1), 4225. <https://doi.org/10.1038/s41598-018-22504-5>.
- Dixit, Y., Hodell, D.A., Petrie, C.A., 2014. Abrupt weakening of the summer monsoon in northwest India ~4100 yr ago. *Geology* 42 (4), 339–342. <https://doi.org/10.1130/G35236.1>.
- Dodia, R., Agrawal, D.P., Vora, A.B., 1984. New pollen data from the Kashmir bogs. In: Whyte, R.O. (Ed.), *The Evolution of the East Asian Environment*, 2. Centre of Asian Studies, University of Hong Kong, pp. 569–573.
- Dutt, S., Gupta, A.K., Clemens, S.C., Cheng, H., Singh, R.K., Kathayat, G., Edwards, R.L., 2015. Abrupt changes in Indian summer monsoon strength during 33,800 to 5500 years. *B.P. Geophys. Res. Lett.* 42 (13), 5526–5532. <https://doi.org/10.1002/2015GL064015>.
- Giesche, A., Staubwasser, M., Petrie, C.A., Hodell, D.A., 2019. Indian winter and summer monsoon strength over the 4.2 ka BP event in foraminifer isotope records from the Indus River delta in the Arabian Sea. *Clim. Past* 15 (1), 73–90. <https://doi.org/10.5194/cp-15-73-2019>.
- Govil, P., Naidu, P.D., 2011. Variations of Indian monsoon precipitation during the last 32 kyr reflected in the surface hydrography of the Western Bay of Bengal. *Quat. Sci. Rev.* 30 (27–28), 3871–3879. <https://doi.org/10.1016/j.quascirev.2011.10.004>.
- Gower, D.J., Agarwal, I., Karanth, K.P., Datta-Roy, A., Giri, V.B., Wilkinson, M., San Mauro, D., 2016. The role of wet-zone fragmentation in shaping biodiversity patterns in peninsular India: insights from the caecilian amphibian *Gegeneophis*. *J. Biogeogr.* 43 (6), 1091–1102. <https://doi.org/10.1111/jbi.12710>.
- Gupta, A.K., Anderson, D.M., Overpeck, J.T., 2003. Abrupt changes in the Asian southwest monsoon during the Holocene and their links to the North Atlantic Ocean. *Nature* 421 (6921), 354–357. <https://doi.org/10.1038/nature01340>.
- Hofmann, S., Rödder, D., Schmidt, J., Flecks, M., Jablonski, D., Dubois, A., Ohler, A., Baniya, C.B., Vershinin, V., Litvinchuk, S.N., Dufresnes, C., 2026. Diversification and historical biogeography of the Himalayan toad (*Duttaphrynus himalayanus*). *Palaeogeogr. Palaeoclimatol. Palaeoecol.* 683, 113473. <https://doi.org/10.1016/j.palaeo.2025.113473>.
- Jones, S.C., 2012. Local- and regional-scale impacts of the ~74 ka Toba supervolcanic eruption on hominin populations and habitats in India. *Quat. Int.* 258, 100–118. <https://doi.org/10.1016/j.quaint.2011.09.017>.
- Joshi, J., Karanth, K.P., 2013. Did southern Western Ghats of peninsular India serve as refugia for its endemic biota during the Cretaceous volcanism? *Ecol. Evol.* 3 (10), 3275–3282. <https://doi.org/10.1002/eece3.603>.
- Kamei, R.G., San Mauro, D., Gower, D.J., Van Bocxlaer, I., Sherratt, E., Thomas, A., Babu, S., Bossuyt, F., Wilkinson, M., Biju, S.D., 2012. Discovery of a new family of amphibians from northeast India with ancient links to Africa. *Proc. Royal Soc. B Biol. Sci.* 279 (1737), 2396–2401. <https://doi.org/10.1098/rspb.2012.0150>.
- Kathayat, G., Cheng, H., Sinha, A., Berkelhammer, M., Zhang, H., Duan, P., Li, H., Li, X., Ning, Y., Edwards, R.L., 2018. Evaluating the timing and structure of the 4.2 ka event in the Indian summer monsoon domain from an annually resolved speleothem record from Northeast India. *Clim. Past* 14 (12), 1869–1879. <https://doi.org/10.5194/cp-14-1869-2018>.
- Kathayat, G., Cheng, H., Sinha, A., Yi, L., Li, X., Zhang, H., Li, H., Ning, Y., Edwards, R.L., 2017. The Indian monsoon variability and civilization changes in the Indian subcontinent. *Sci. Adv.* 3 (12), e1701296. <https://doi.org/10.1126/sciadv.1701296>.
- Kathayat, G., Sinha, A., Breitenbach, S.F.M., Tan, L., Spötl, C., Li, H., Dong, X., Zhang, H., Ning, Y., Allan, R.J., Damaraju, V.R., Sinha, A.K., Edwards, R.L., Cheng, H., 2022. Protracted Indian monsoon droughts of the past millennium and their societal impacts. *Proc. Natl. Acad. Sci.* 119 (39), e2207487119. <https://doi.org/10.1073/pnas.2207487119>.
- Kumar, K.S., Chandrika, S.K., George, S., 2020. Genetic structure and demographic history of *Indirana semipalmata*, an endemic frog species of the Western Ghats, India. *Mitochondrial DNA Part A* 31 (8), 365–378. <https://doi.org/10.1080/24701394.2020.1830077>.
- Lajmi, A., Karanth, K.P., 2020. Eocene–Oligocene cooling and the diversification of *Hemidactylus* geckos in Peninsular India. *Mol. Phylogenet. Evol.* 142, 106637. <https://doi.org/10.1016/j.ympev.2019.106637>.
- Li, J.T., Li, Y., Klaus, S., Rao, D.Q., Hillis, D.M., Zhang, Y.P., 2013. Diversification of rhacophorid frogs provides evidence for accelerated faunal exchange between India and Eurasia during the Oligocene. *Proc. Natl. Acad. Sci.* 110 (9), 3441–3446. <https://doi.org/10.1073/pnas.1300881111>.
- Loarie, S.R., Duffy, P.B., Hamilton, H., Asner, G.P., Field, C.B., Ackerly, D.D., 2009. The velocity of climate change. *Nature* 462 (7276), 1052–1055. <https://doi.org/10.1038/nature08649>.
- Luedtke, J.A., Chanson, J., Neam, K., Hobin, L., Maciel, A.O., Catenazzi, A., Borzée, A., Hamidy, A., Aowphol, A., Jean, A., Sosa-Bartuano, A., Fong, G.A., de Silva, A., Fouquet, A., Angulo, A., Kidov, A.A., et al., 2023. Ongoing declines for the world's amphibians in the face of emerging threats. *Nature* 622 (7982), 308–314. <https://doi.org/10.1038/s41586-023-06578-4>.
- Lyman, R.L., 1994. *Vertebrate Taphonomy*. Cambridge University Press, Cambridge.
- Meegaskumbura, M., Bossuyt, F., Pethiyagoda, R., Manamendra-Arachchi, K., Bahir, M., Milinkovitch, M.C., Schneider, C.J., 2002. Sri Lanka: an amphibian hotspot. *Science* 298 (5592), 379. <https://doi.org/10.1126/science.298.5592.379>.
- Meegaskumbura, M., Seneviratne, G., Manamendra-Arachchi, K., Pethiyagoda, R., Hanken, J., Schneider, C.J., 2019. Diversification of shrub frogs (Rhacophoridae, *Pseudophilautus*) in Sri Lanka. Timing and geographic context. *Mol. Phylogenet. Evol.* 132, 14–24. <https://doi.org/10.1016/j.ympev.2018.11.004>.
- Myers, N., Mittermeier, R.A., Mittermeier, C.G., da Fonseca, G.A.B., Kent, J., 2000. Biodiversity hotspots for conservation priorities. *Nature* 403 (6772), 853–858. <https://doi.org/10.1038/35002501>.
- Patnaik, R., 2003. Reconstruction of Upper Siwalik palaeoecology and palaeoclimatology using microfossil palaeocommunities. *Palaeogeogr. Palaeoclimatol. Palaeoecol.* 197 (1–2), 133–150. [https://doi.org/10.1016/S0031-0182\(03\)00390-0](https://doi.org/10.1016/S0031-0182(03)00390-0).
- Patnaik, R., Badam, G.L., Murty, M.L.K., 2008. Additional vertebrate remains from one of the Late Pleistocene–Holocene Kurnool Caves (Muchchatla Chintamanu Gavi) of South India. *Quat. Int.* 192 (1), 43–51. <https://doi.org/10.1016/j.quaint.2007.06.018>.
- Patnaik, R., Chauhan, P.R., Rao, M.R., Blackwell, B.A.B., Skinner, A.R., Sahni, A., Chauhan, M.S., Khan, H.S., 2009. New geochronological, paleoclimatological, and archaeological data from the Narmada Valley hominin locality, central India. *J. Hum. Evol.* 56 (2), 114–133. <https://doi.org/10.1016/j.jhevol.2008.08.023>.
- Petraglia, M., Korisettar, R., Boivin, N., Clarkson, C., Ditchfield, P., Jones, S., Koshy, J., Lahr, M.M., Oppenheimer, C., Pyle, D., Roberts, R., Schwenninger, J.-L., Arnold, L., White, K., 2007. Middle Paleolithic assemblages from the Indian subcontinent before and after the Toba super-eruption. *Science* 317 (5834), 114–116. <https://doi.org/10.1126/science.1141564>.

- Prasad, S., Enzel, Y., 2006. Holocene paleoclimates of India. *Quat. Res.* 66 (3), 442–453. <https://doi.org/10.1016/j.yqres.2006.05.008>.
- Rio, J.P., Mannion, P.D., 2024. Osteology of *Crocodylus palaeindicus* from the late Miocene–Pleistocene of South Asia and the phylogenetic relationships of crocodylids. *J. Syst. Palaeontol.* 22 (1), 2313133. <https://doi.org/10.1080/14772019.2024.2313133>.
- Robin, V.V., Sinha, A., Ramakrishnan, U., 2010. Ancient geographical gaps and paleoclimate shape the phylogeography of an endemic bird in the sky islands of southern India. *PLoS One* 5 (10), e13321. <https://doi.org/10.1371/journal.pone.0013321>.
- Sandeep, K., Shankar, R., Kumar, A., Yadava, M.G., Ramesh, R., Jani, R.A., Weijian, Z., Xuefeng, L., 2017. A multi-proxy lake sediment record of Indian summer monsoon variability during the Holocene in southern India. *Palaeogeogr. Palaeoclimatol. Palaeoecol.* 476, 1–14. <https://doi.org/10.1016/j.palaeo.2017.03.021>.
- Saraswat, R., Lea, D.W., Nigam, R., Mackensen, A., Naik, D.K., 2013. Deglaciation in the tropical Indian Ocean driven by interplay between the regional monsoon and global teleconnections. *Earth Planet Sci. Lett.* 375, 166–175. <https://doi.org/10.1016/j.epsl.2013.05.022>.
- Sharma, S., Joachimski, M., Sharma, M., Tobschall, H.J., Singh, I.B., Sharma, C., Chauhan, M.S., Morgenroth, G., 2004. Lateglacial and Holocene environmental changes in Ganga plain, northern India. *Quat. Sci. Rev.* 23 (1–2), 145–159. <https://doi.org/10.1016/j.quascirev.2003.10.005>.
- Singh, G., 1963. A preliminary survey of the post-glacial vegetational history of the Kashmir Valley. *Palaeobot* 12 (1), 73–108.
- Singh, G., Agrawal, D.P., 1976. Radiocarbon evidence for deglaciation in north-western Himalaya, India. *Nature* 260 (5548), 232. <https://doi.org/10.1038/260232a0>.
- Sinha, A., Berkelhammer, M., Stott, L., Mudelsee, M., Cheng, H., Biswas, J., 2011. The leading mode of Indian Summer Monsoon precipitation variability during the last millennium. *Geophys. Res. Lett.* 38 (15), L15703. <https://doi.org/10.1029/2011GL047713>.
- Sinha, A., Cannariato, K.G., Stott, L.D., Cheng, H., Edwards, R.L., Yadava, M.G., Ramesh, R., Singh, I.B., 2007. A 900-year (600 to 1500 A.D.) record of the Indian summer monsoon precipitation from the core monsoon zone of India. *Geophys. Res. Lett.* 34 (16), L16707. <https://doi.org/10.1029/2007GL030431>.
- Sinha, A., Kathayat, G., Cheng, H., Breitenbach, S.F.M., Berkelhammer, M., Mudelsee, M., Biswas, J., Edwards, R.L., 2015. Trends and oscillations in the Indian summer monsoon rainfall over the last two millennia. *Nat. Commun.* 6, 6309. <https://doi.org/10.1038/ncomms7309>.
- Sinha, A., Kathayat, G., Weiss, H., Li, H., Cheng, H., Reuter, J., Schneider, A.W., Berkelhammer, M., Adali, S.F., Stott, L.D., Edwards, R.L., 2018. Role of climate in the rise and fall of the Neo-Assyrian Empire. *Sci. Adv.* 4 (11), eaat9871. <https://doi.org/10.1126/sciadv.aax6656>.
- Srinivasulu, A., Srinivasulu, B., Srinivasulu, C., 2021. Ecological niche modelling for the conservation of endemic threatened squamates (lizards and snakes) in the Western ghats. *Glob. Ecol. Conserv.* 28, e01700. <https://doi.org/10.1016/j.gecco.2021.e01700>.
- Staubwasser, M., Sirocko, F., Grootes, P.M., Segl, M., 2003. Climate change at the 4.2 ka BP termination of the Indus valley civilization and Holocene south Asian monsoon variability. *Geophys. Res. Lett.* 30 (8), 1425. <https://doi.org/10.1029/2002GL016822>.
- Subba, B., Sen, S., Ravikanth, G., Nobis, M.P., 2018. Direct modelling of limited migration improves projected distributions of Himalayan amphibians under climate change. *Biol. Conserv.* 227, 352–360. <https://doi.org/10.1016/j.biocon.2018.09.035>.
- Turvey, S.T., Sathe, V., Crees, J.J., Jukar, A.M., Chakraborty, P., Lister, A.M., 2021. Late Quaternary megafaunal extinctions in India: how much do we know? *Quat. Sci. Rev.* 252, 106740. <https://doi.org/10.1016/j.quascirev.2020.106740>.
- Van Bocxlaer, I., Biju, S.D., Loader, S.P., Bossuyt, F., 2009. Toad radiation reveals into-India dispersal as a source of endemism in the Western Ghats–Sri Lanka biodiversity hotspot. *BMC Evol. Biol.* 9, 131. <https://doi.org/10.1186/1471-2148-9-131>.
- Van Bocxlaer, I., Roelants, K., Biju, S.D., Nagaraju, J., Bossuyt, F., 2006. Late Cretaceous vicariance in Gondwanan amphibians. *PLoS One* 1 (1), e74. <https://doi.org/10.1371/journal.pone.0000074>.
- Vijayakumar, S.P., Menezes, R.C., Jayarajan, A., Shanker, K., 2016. Glaciations, gradients, and geography: multiple drivers of diversification of bush frogs in the Western Ghats Escarpment. *Proc. Royal Soc. B Biol. Sci.* 283 (1836), 20161011. <https://doi.org/10.1098/rspb.2016.1011>.
- Yost, C.L., Jackson, L.J., Stone, J.R., Cohen, A.S., 2018. Subdecadal phytolith and charcoal records from Lake Malawi, East Africa imply minimal effects on human evolution from the ~74 ka Toba supereruption. *J. Hum. Evol.* 116, 75–94. <https://doi.org/10.1016/j.jhevol.2017.11.005>.